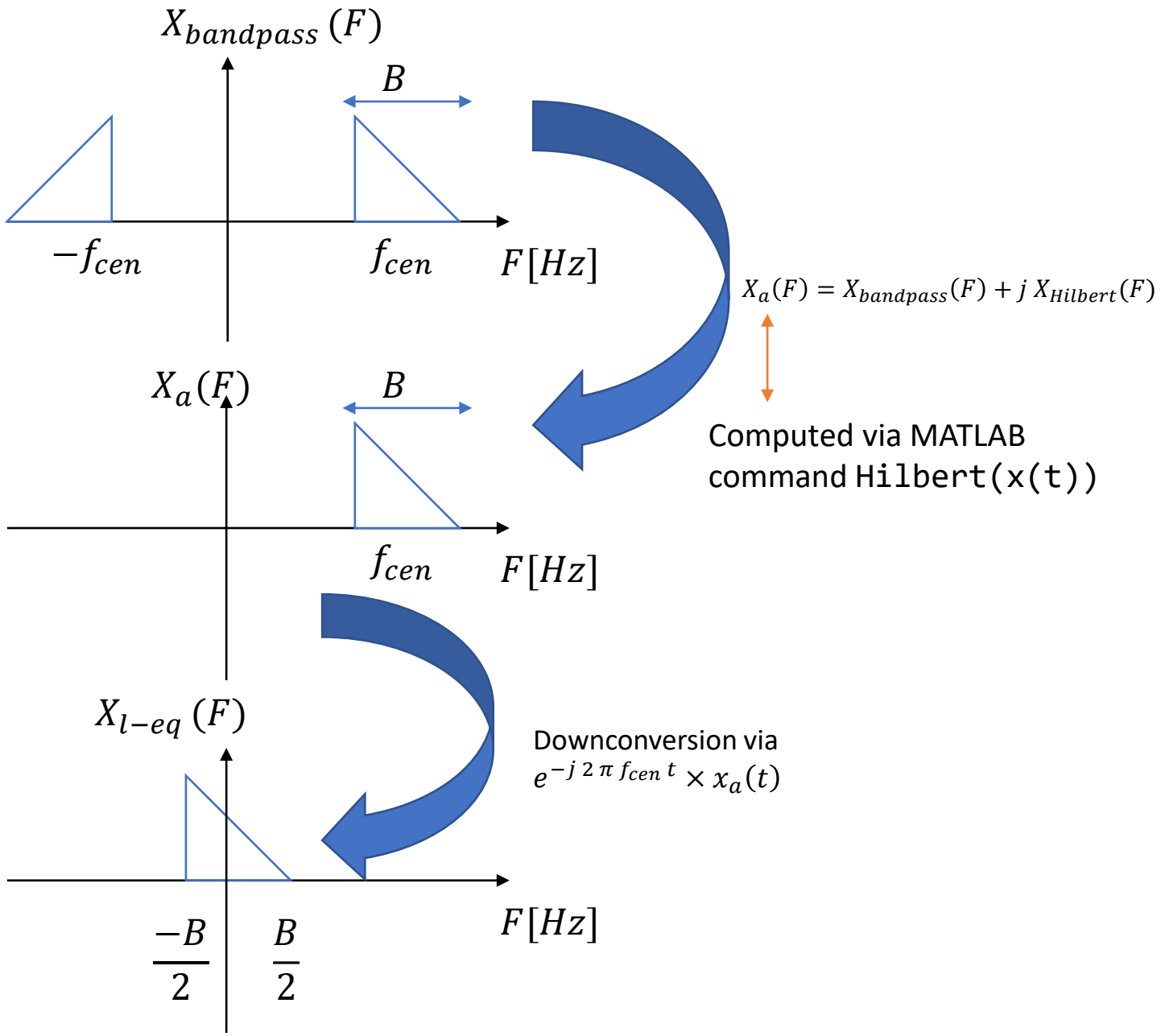


Summary of Formula

$$x_{l-eq\,(t)} = x_I(t) + j\,x_Q(t)$$
$$= Amp(t)\,\exp[\,j\,\phi(t)]$$

$$Amp(t) = \left[x_I^2(t) + x_Q^2(t)\right]^{0.5}$$
$$\phi(t) = \tan^{-1} x_Q(t)/x_I(t)$$

$$x_I(t) = Amp(t)\cos\phi(t)$$
$$x_Q(t) = Amp(t)\,sin\,\phi(t)$$